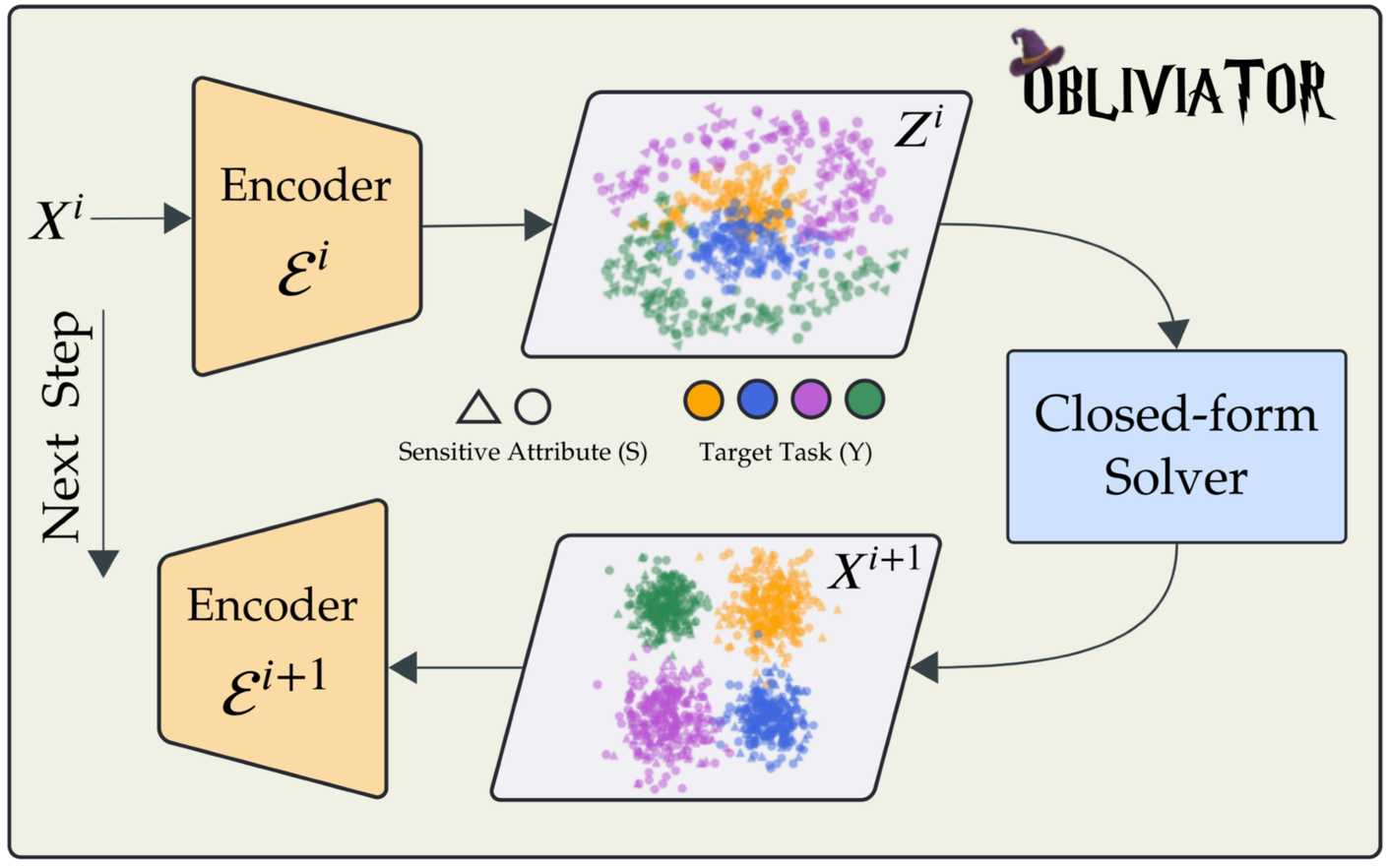
How Do We Solve This Optimization ?



Step One

Erase and **Preserve** via guarding function $Z_\theta^i = \varepsilon^i(X^i; \boldsymbol{\theta}^i)$

$$\inf_{\theta} \quad \mathbf{HSIC}(Z_\theta, S) - \left[\mathbf{HSIC}(Z_\theta^i, Y) + \mathbf{HSIC}(Z_\theta^i, X) + \mathbf{HSIC}(Z_\theta^i, X^i)\right]$$

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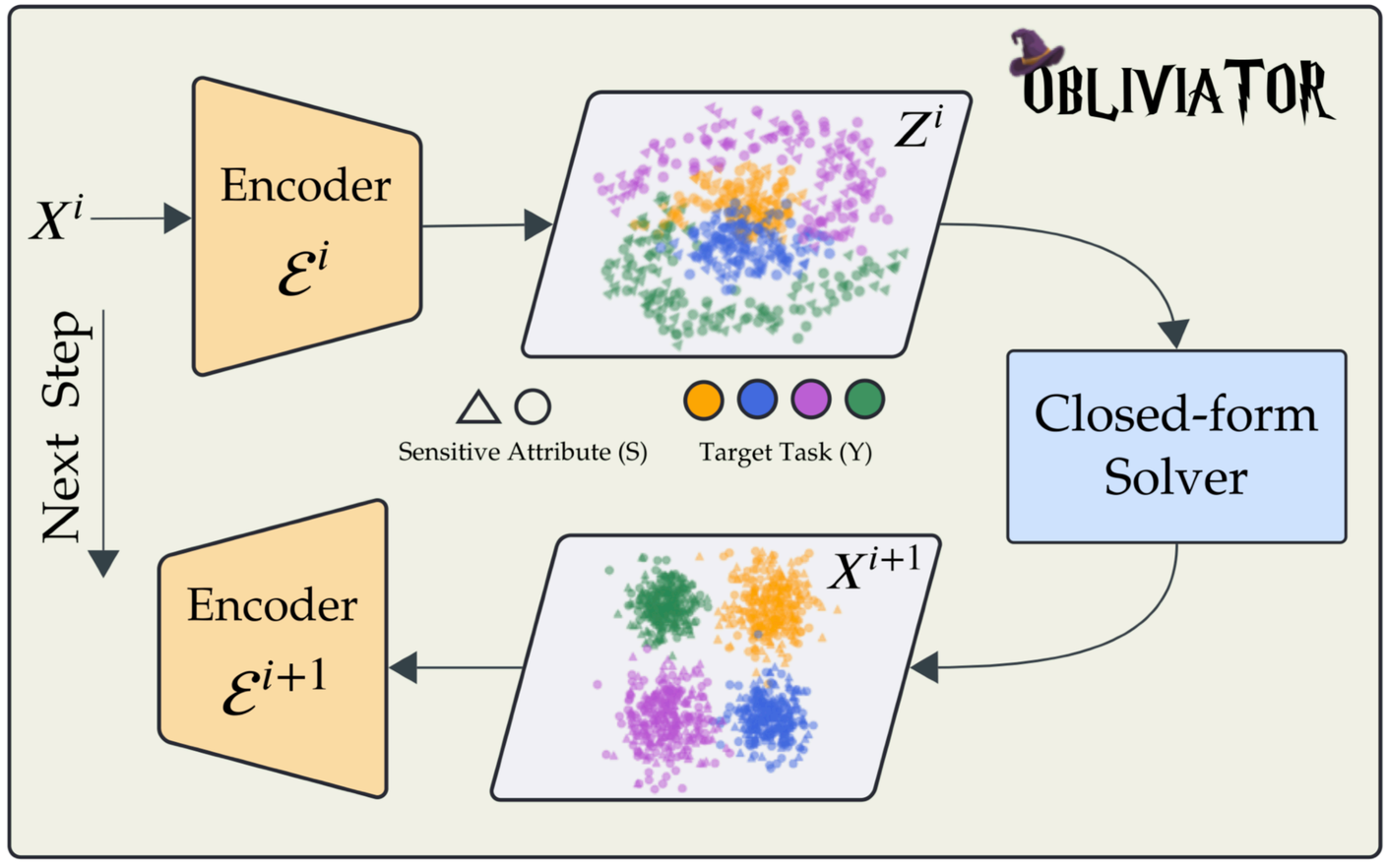

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More Information-Preserving Loss